

CANDIDATE
NAME

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NUMBER

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BIOLOGY

9700/53

Paper 5 Planning, Analysis and Evaluation

October/November 2017

1 hour 15 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **8** printed pages and **4** blank pages.

- 1 The enzyme urease catalyses the breakdown of urea into carbonate ions and ammonium ions as shown in Fig. 1.1.

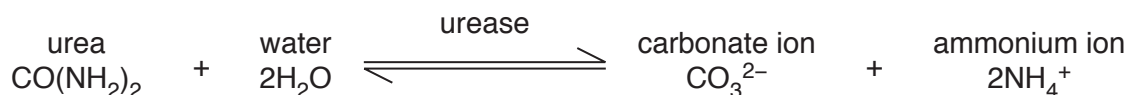


Fig. 1.1

A solution of urea does not conduct electricity, but a solution of the two ions does conduct electricity.

An increase in ions increases conductivity, so the rate of reaction is proportional to the conductivity.

A meter measures conductivity in microsiemens per centimetre ($\mu\text{S cm}^{-1}$). The conductivity meter also records temperature.

K_m shows the affinity of an enzyme for its substrate. The lower the K_m the greater the affinity. A student carried out an investigation to find the K_m of the enzyme urease at different temperatures.

The student:

- made 500 cm^3 of 0.2 mol dm^{-3} solution of urea, molar mass 60.1 g mol^{-1}
- used serial dilution to make a total of five urea solutions from 0.2 mol dm^{-3} stock solution
- used 1 g per 10 cm^3 urease solution
- used a conductivity meter to measure the initial rate of reaction at each temperature.

Fig. 1.2 shows the experimental set-up during the course of a reaction.

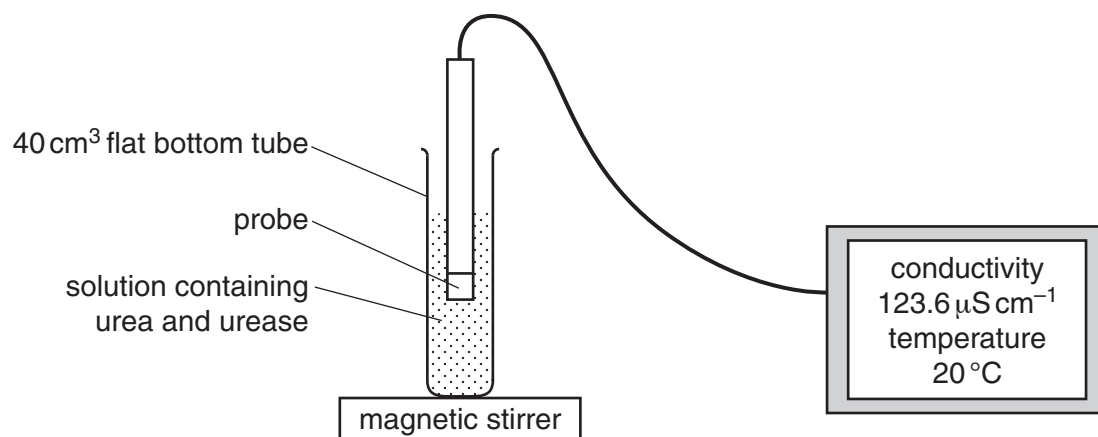


Fig. 1.2

- (a) (i) Describe how the student could make 500 cm^3 of 0.2 mol dm^{-3} solution of urea.

.....

.....

.....

.....

..... [2]

- (ii) Describe how the student made a further **four** solutions of urea by **serial dilution**.

.....

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.....

.....

..... [2]

- (b) (i) Identify the independent and dependent variables in this investigation.

independent

dependent [2]

- (ii) Suggest a suitable control for this investigation.

.....

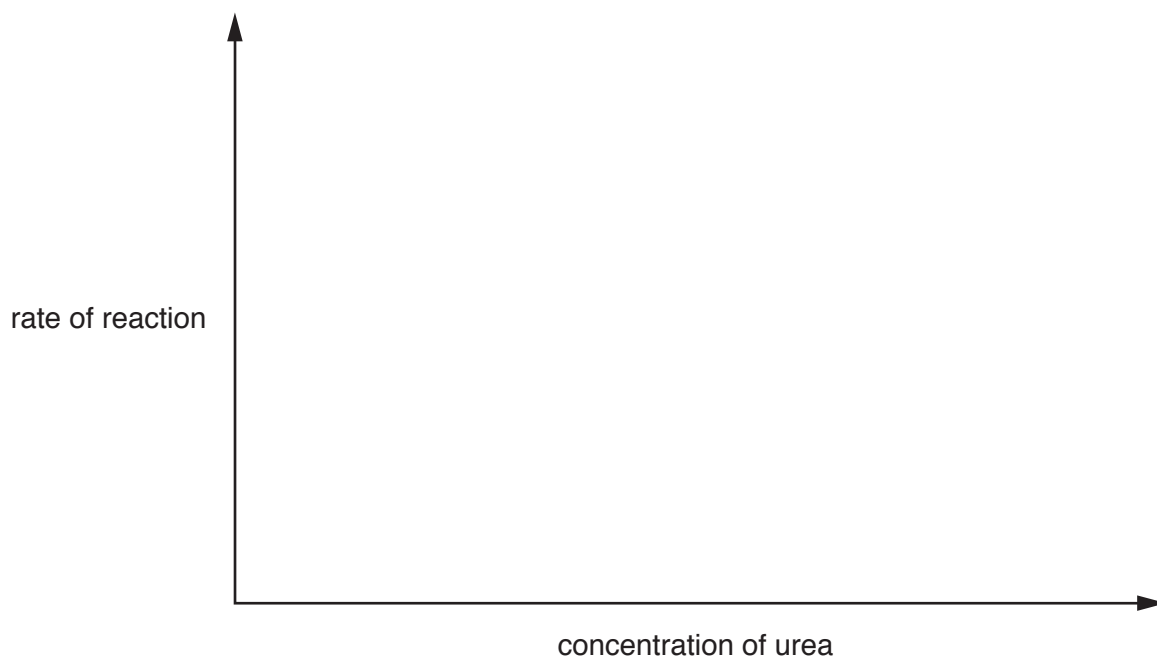
..... [1]

Your method should be set out in a logical way and be detailed enough to let another person follow it. You should **not** include details of how to prepare the urea or urease solutions.

[6]

- (c) (i) Use the axes below to sketch a graph to show the effect of substrate concentration on the initial rate of reaction at **one** temperature.

Indicate on your graph how the student could find the K_m of urease at that temperature.



[4]

- (ii) Table 1.1 shows the results that the student obtained for K_m at different temperatures.

Table 1.1

temperature	A	B	C	D	E
K_m /arbitrary units	1.8	1.3	0.9	0.8	1.2

State which of these temperatures, **A** to **E**, is closest to the likely optimum temperature of urease.

Explain your answer.

.....

.....

.....

.....

..... [2]

[Total: 19]

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2 Barley is often grown in soils that are high in salt (sodium chloride).

Researchers tested the salt tolerance of two varieties of barley, **X** and **Y**, by germinating grains in seven different concentrations of salt solution.

- Seven sets of barley grains of each variety were soaked in the different concentrations of salt solution for 12 hours. Each set contained 50 grains.
- Each set of grains was then split into groups of 10 grains. These were placed in Petri dishes on filter paper soaked in the same concentrations of salt solution that were used for soaking the grains.
- Lids were placed on the Petri dishes and the grains were left at 20 °C to germinate.

Fig. 2.1 shows the arrangement of the grains in the Petri dishes.

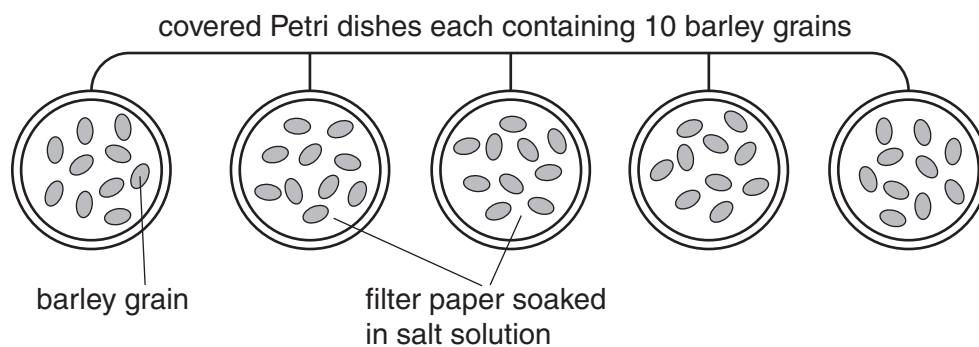


Fig. 2.1

The grains were then observed at 8 hour intervals for a total of five days. The percentage of grains in each set of 50 that had germinated each day was calculated.

The first appearance of the radicle (root) was used to indicate that the grains had germinated.

If no radicle appeared after five days, germination was considered to have failed.

The researchers repeated the whole investigation several times and calculated mean results.

(a) (i) Describe **one** variable that has been standardised in this investigation.

.....
 [1]

(ii) Suggest **one** other variable that should be standardised.

.....
 [1]

concentration of salt solution /mmol dm ⁻³	mean (cumulative) percentage of grains that had germinated each day			
	1		5	
	X	Y	X	Y
0	93.8	92.0	98.0	97.0
10	93.0	90.2	97.4	96.4
20	92.2	87.0	96.6	93.6
30	96.4	90.0	96.4	93.0
40	92.0	90.4	95.4	92.4
50	91.3	91.0	95.0	91.6
60	91.2	90.0	95.0	91.0

-

[2]

- [3]

[3]

- (d) The researchers concluded that variety **X** would be better suited than variety **Y** to grow in salty soil.

Other researchers did not think there was sufficient evidence to make definite conclusions.

- (i) State the evidence in Table 2.1 that supports the conclusion that variety **X** is better suited than variety **Y** to grow in salty soil.

.....
 [1]

- (ii) The researchers decided to carry out further investigations to find out which is the most suitable variety, **X** or **Y**, to grow in salty soils.

Suggest **one** other investigation they could carry out.

.....

 [1]

[Total: 11]

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